

Diabetes Analysis with Bayesian Network

Fundamentals Of AI - Module 3
Professor Paolo Torroni

Sara Tajernia
Mohammad Choupan
Mohammad Amin Mirzakuchaki

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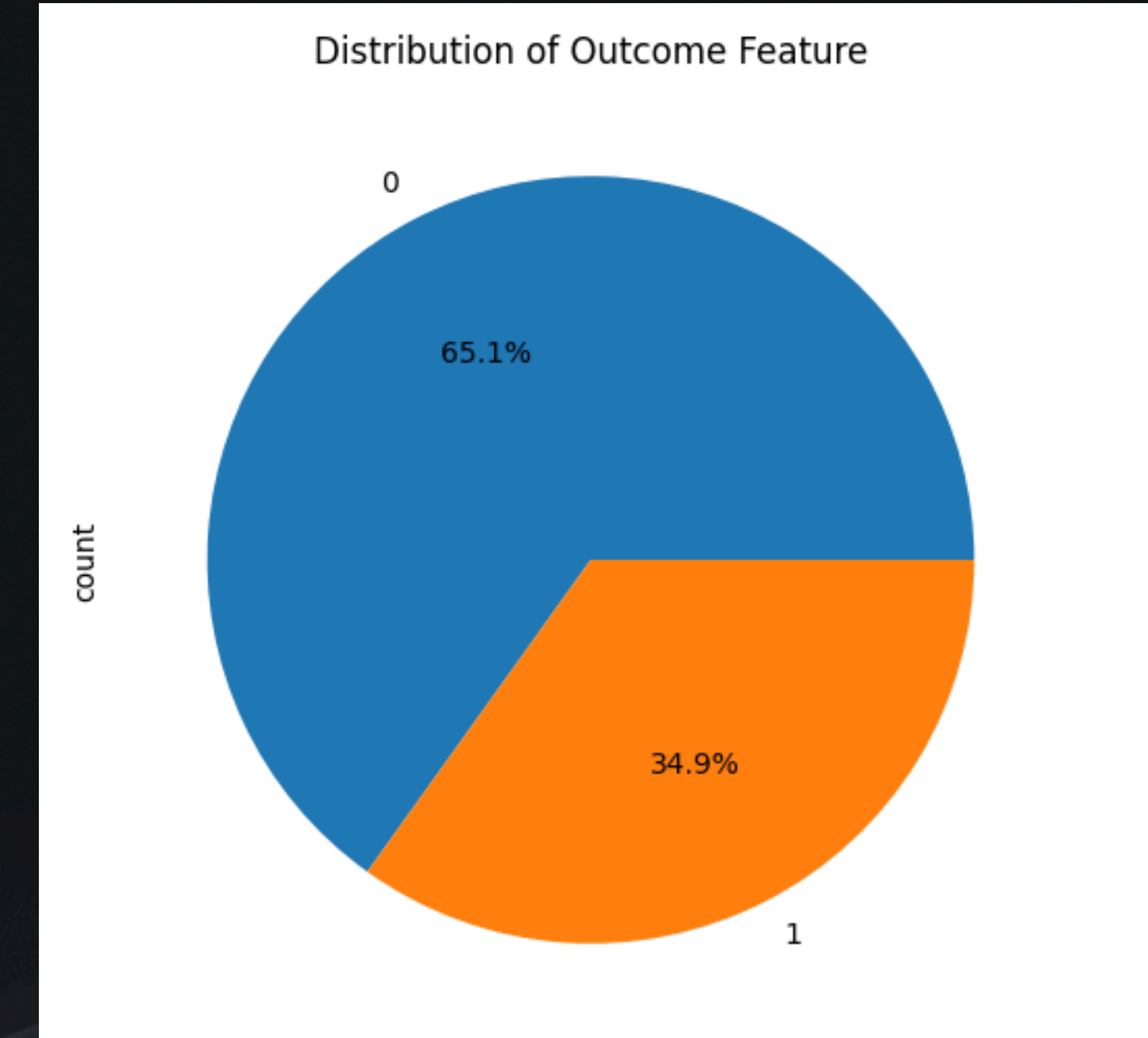
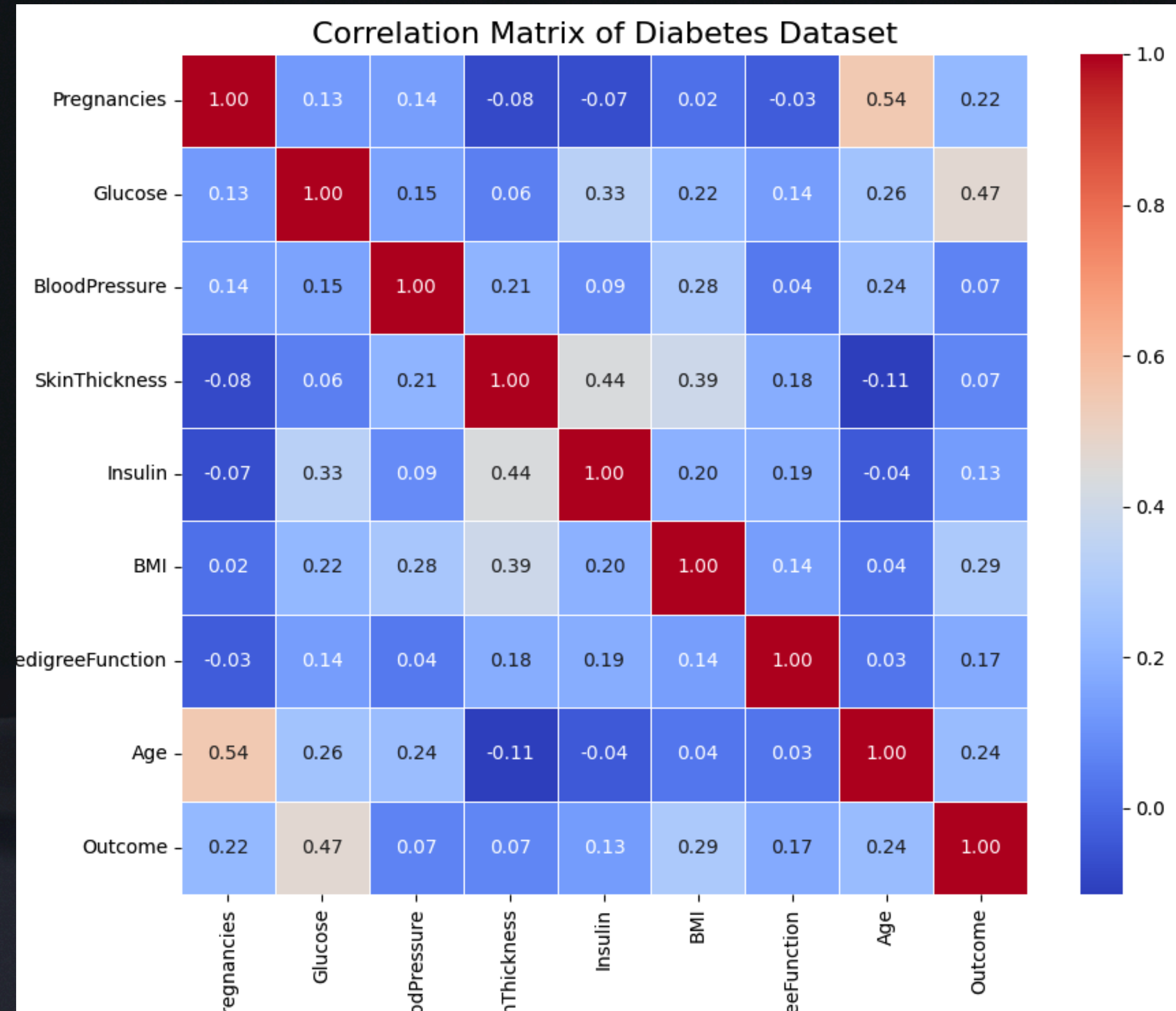
Dataset Overview

<i>Attribute</i>	<i>Description</i>
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration 2 hours after an oral glucose tolerance test
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-hour serum insulin (μ U/ml)
BMI	Body mass index (weight in kg / height in m ²)
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age (years)
Outcome	Class variable (0 or 1) indicating if the patient has diabetes (1) or not (0)

Exploratory Data Analysis

- statistical summary of the dataset
- plotting distributions of features
- checking for missing values
- checking for zero values in features where it is not possible
- scatter plot of each feature vs Outcome
- pair plot of features
- heatmap of correlation matrix

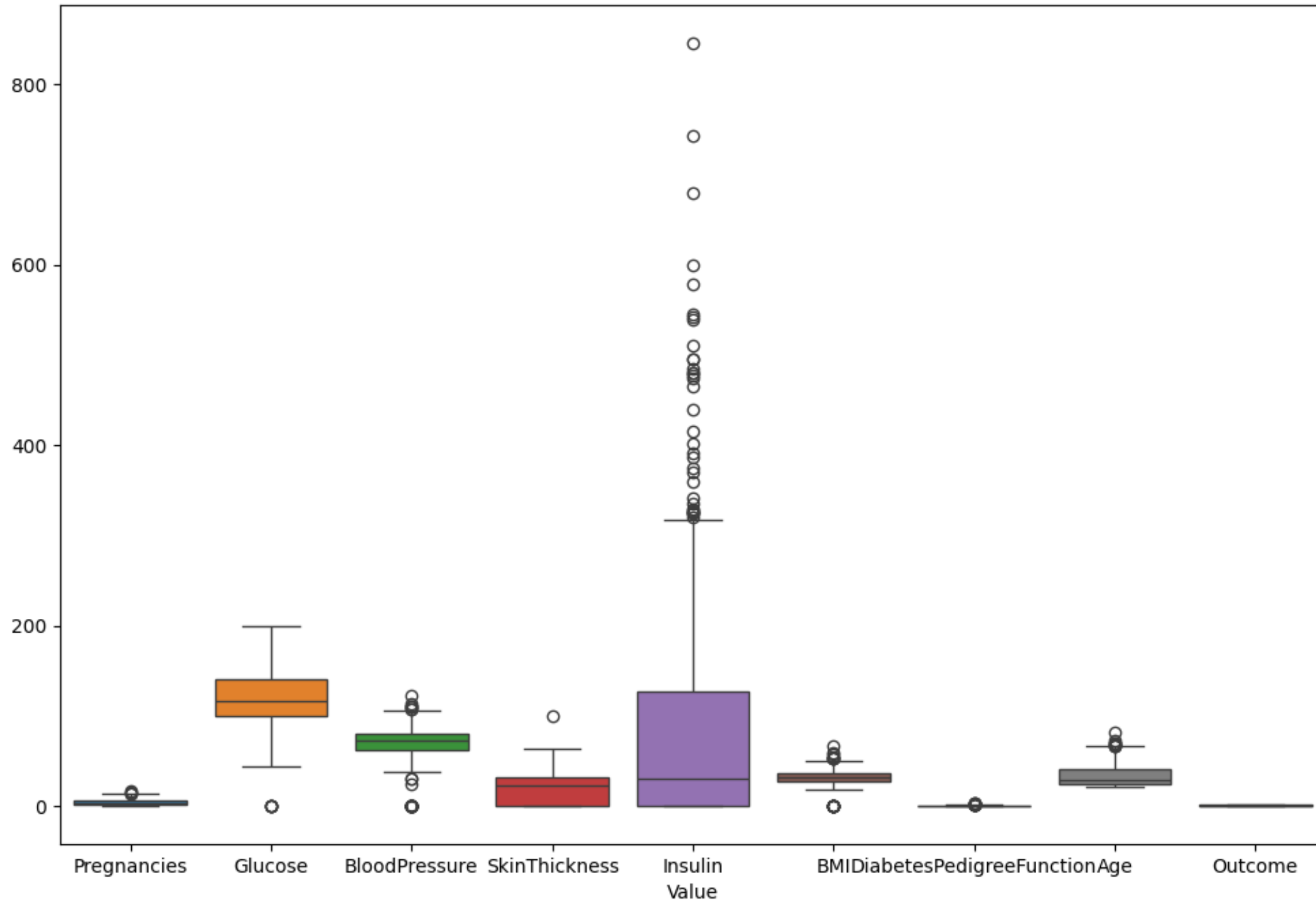
Data Sourcing



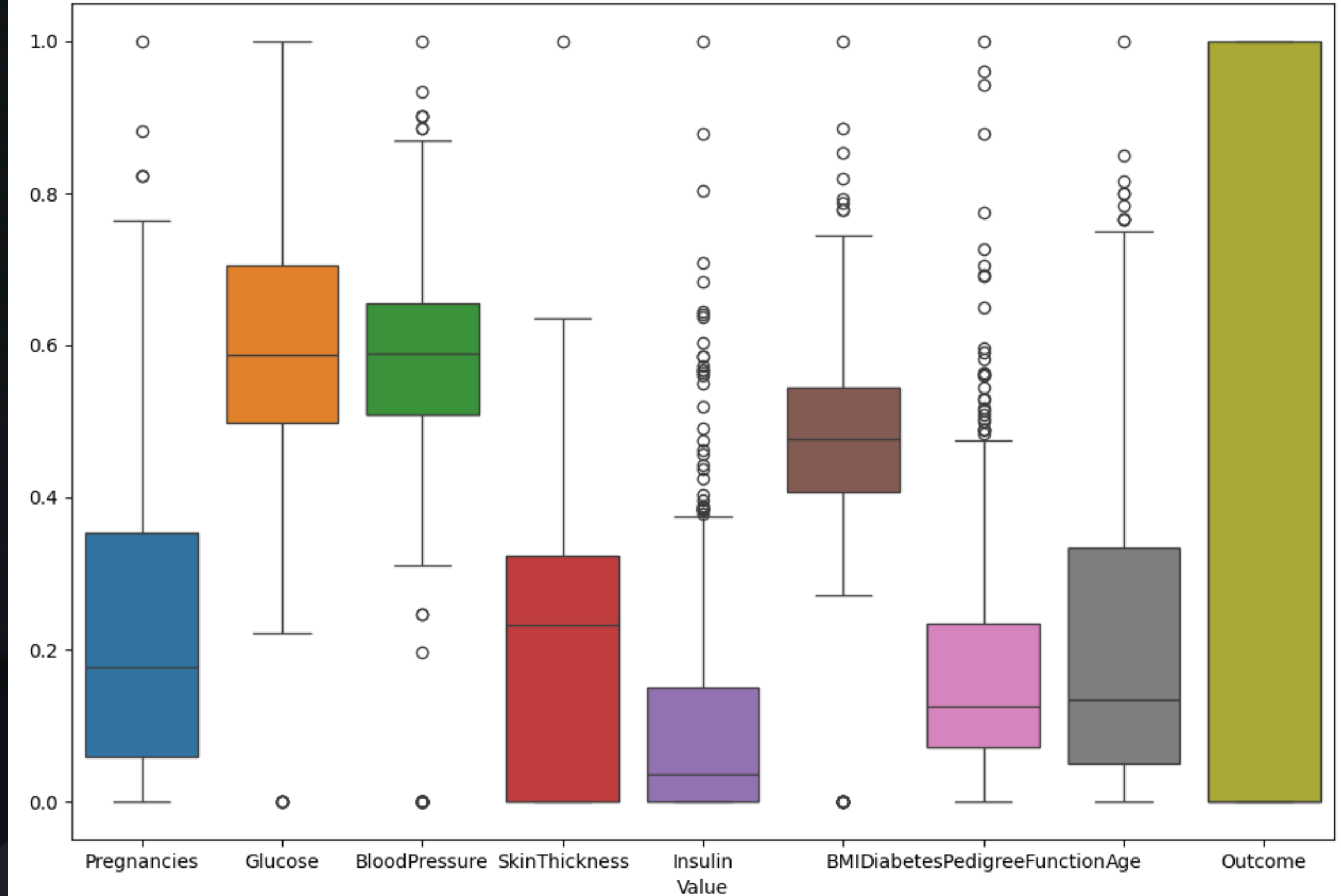
- 768 total instances
- 500 non-diabetic (0), 268 diabetic (1)
- Mix of continuous and categorical features

Normalize (MinMax Scaler)

Box Plot of Each Feature in Diabetes Dataset



Box Plot of Each Feature in Normalized Diabetes Dataset

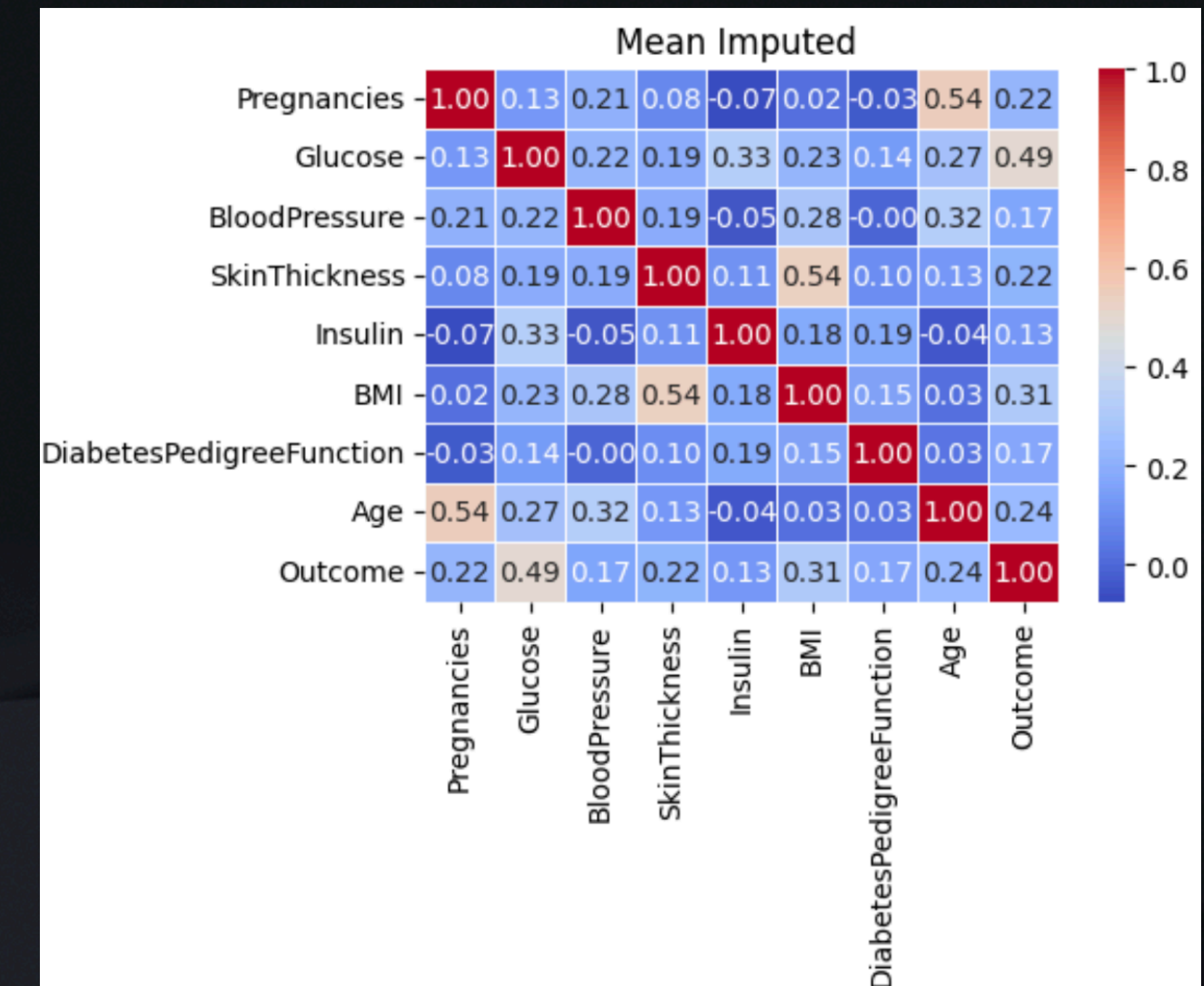
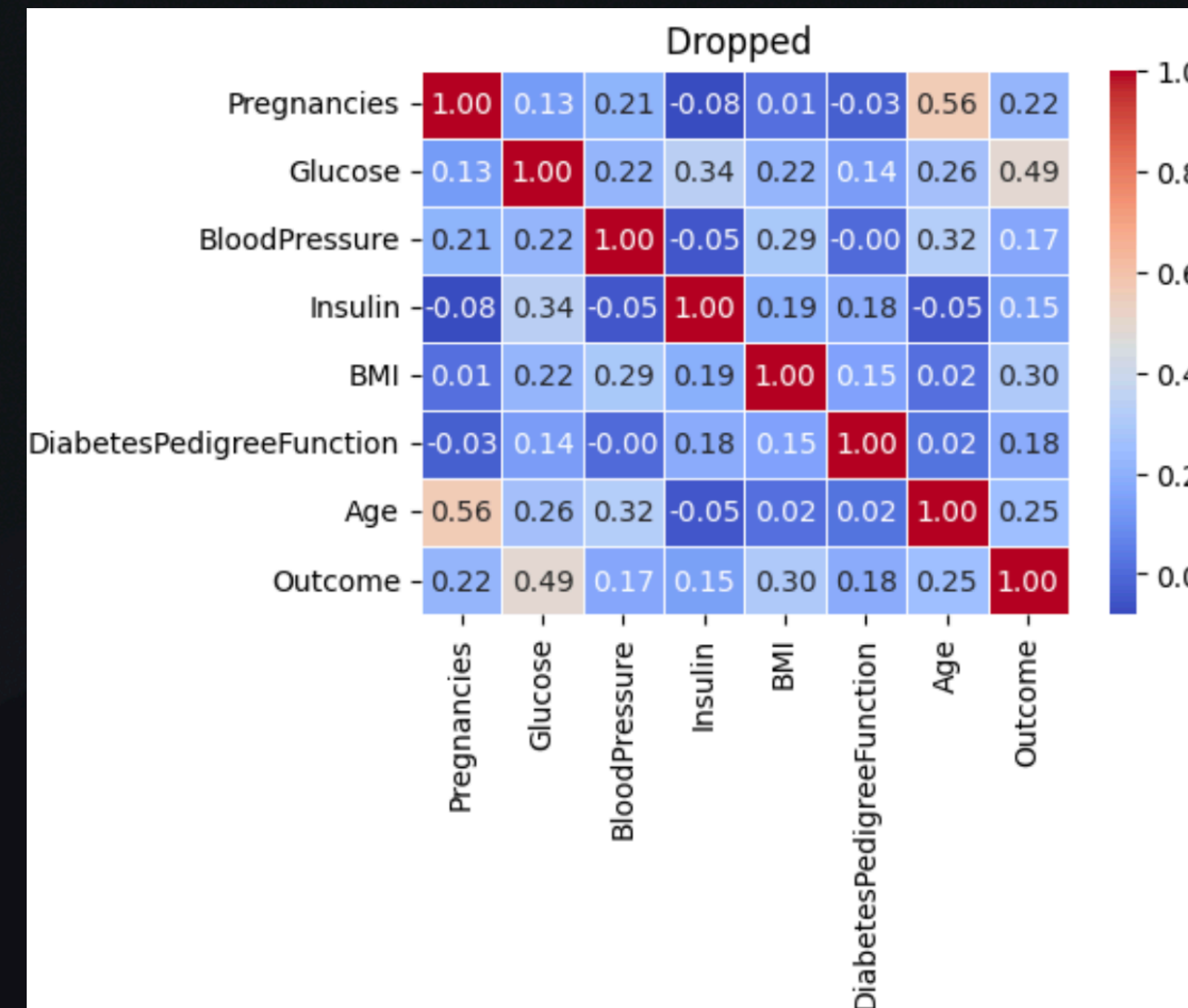
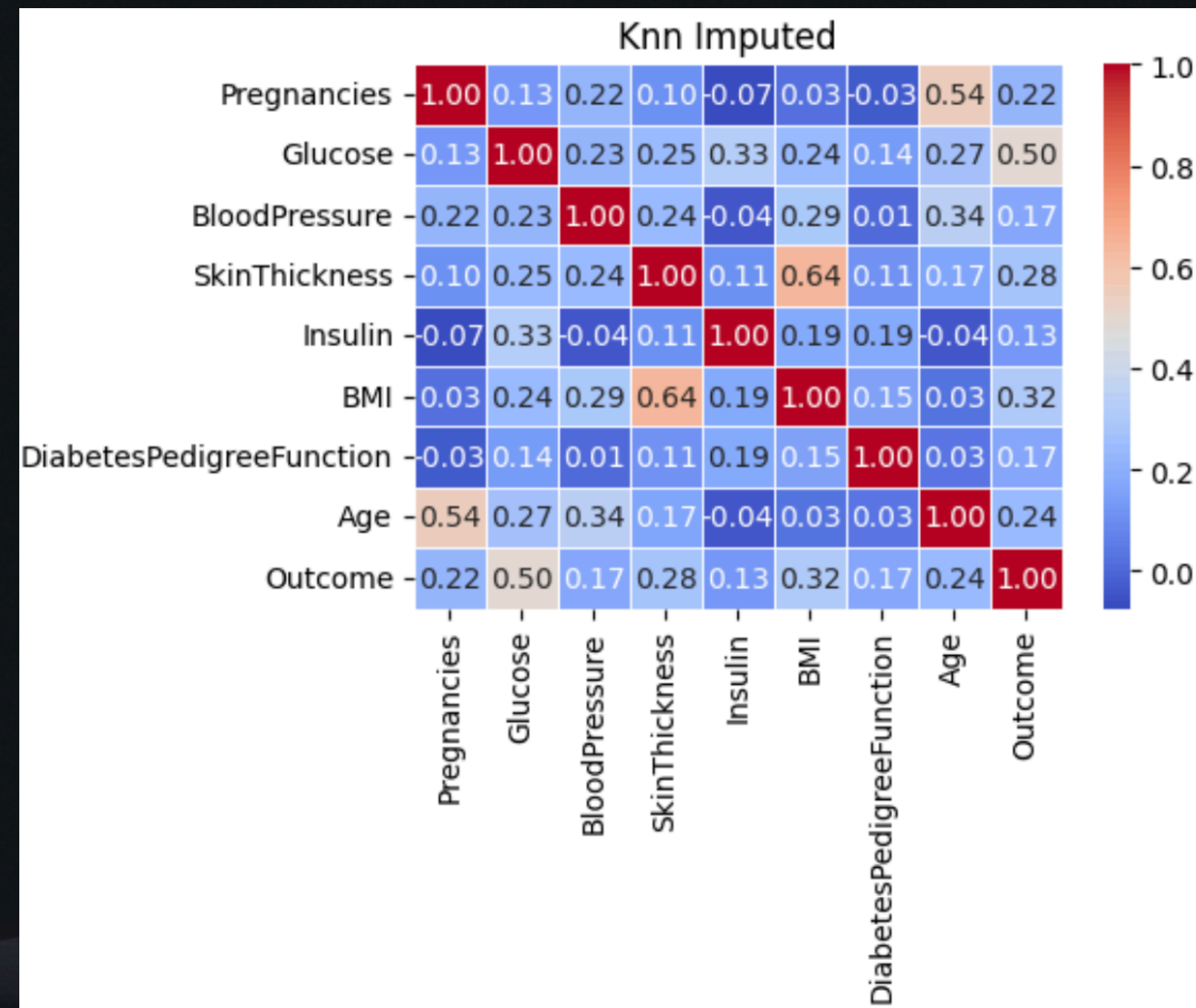


Preprocessing

- *Handle Zero Values*
 - Remove rows with missing values ✓
 - **Remove column with most missing values and remove rows with missing values in other columns ✓ (SkinThickness)**
 - *Impute missing values with mean/median/mode*
 - *Use advanced imputation techniques like KNN imputation or regression imputation*

{'Glucose': 5, 'BloodPressure': 35, 'SkinThickness': 227, 'BMI': 11, 'Age': 0}

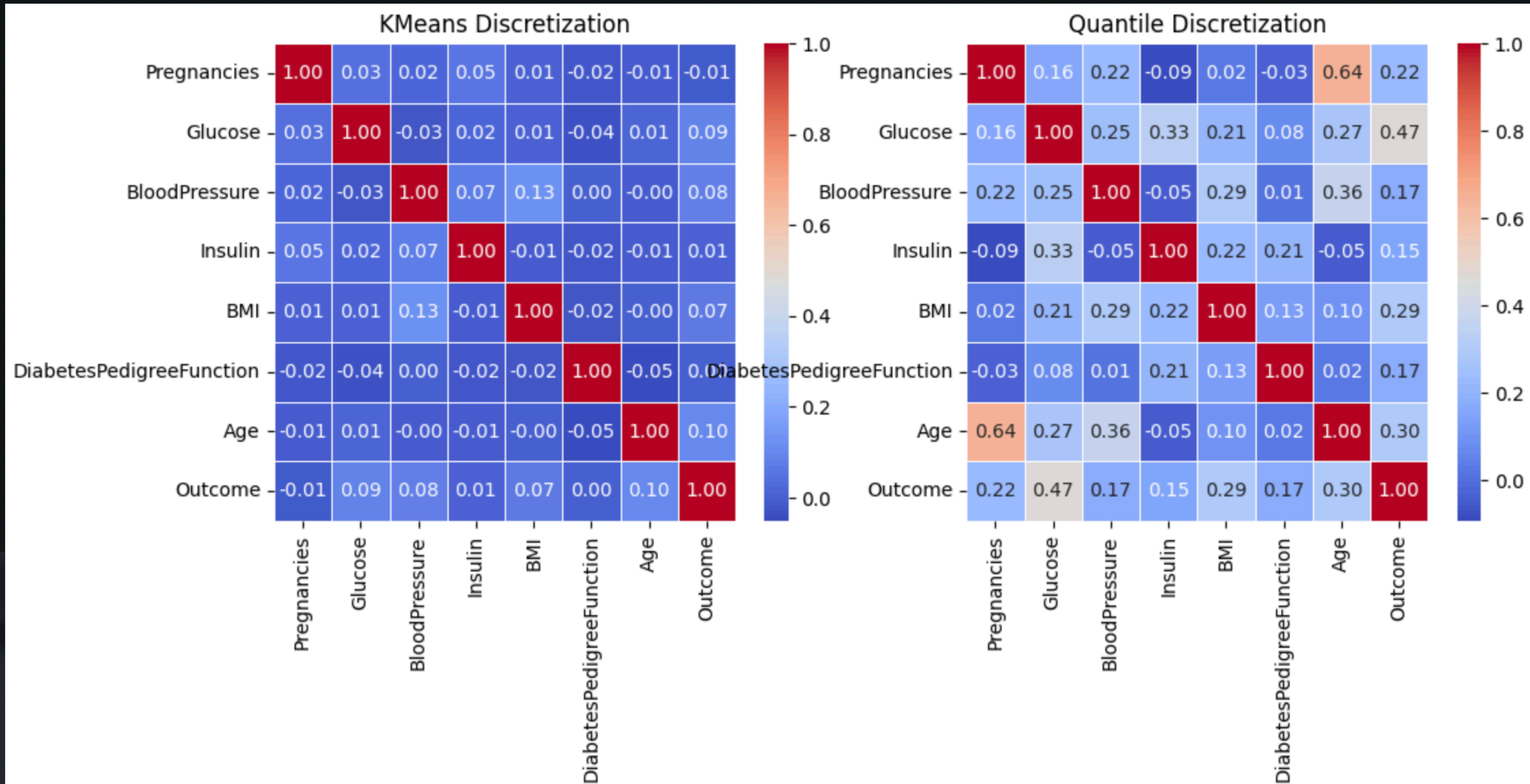
Handle Missing Values



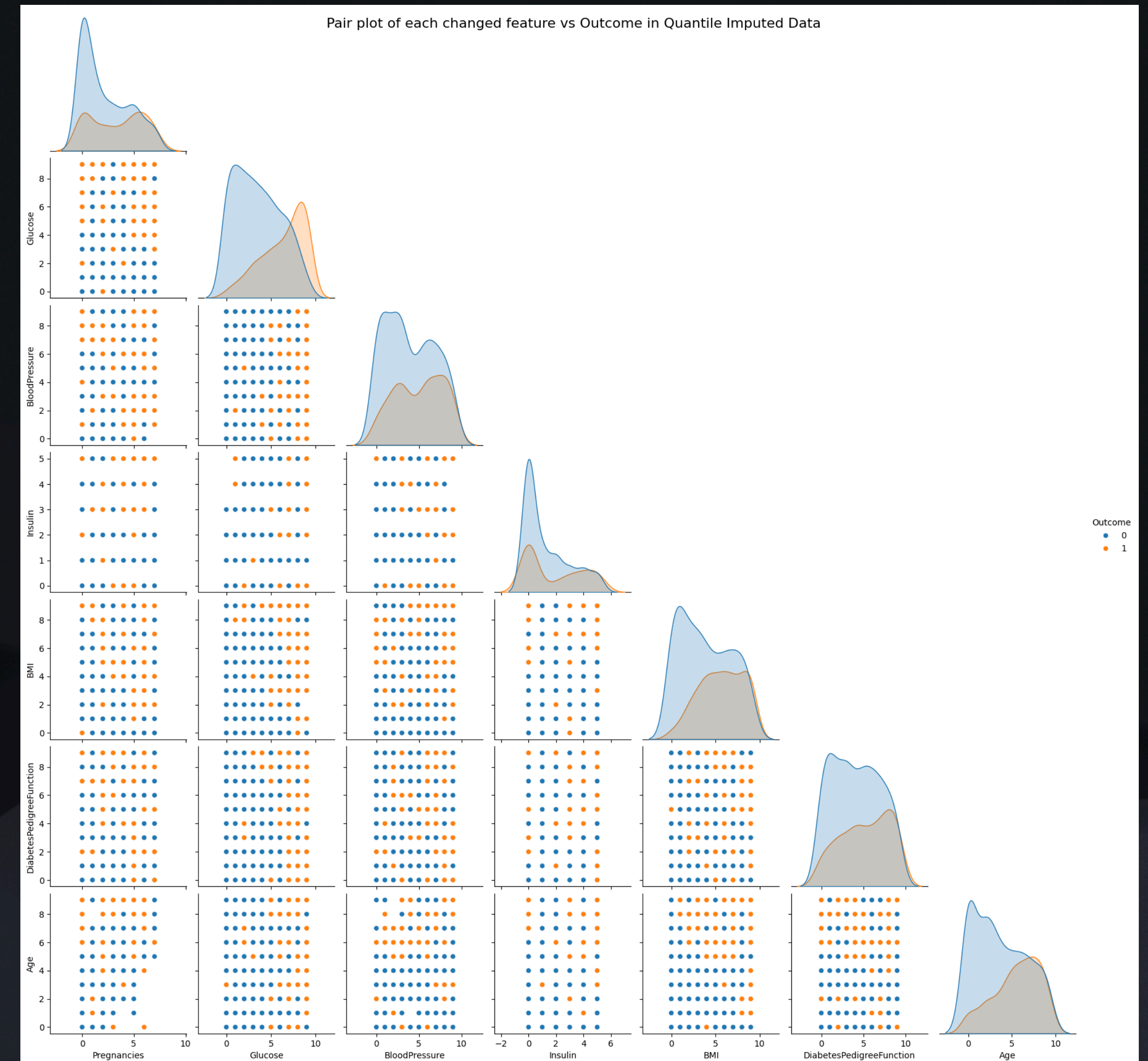
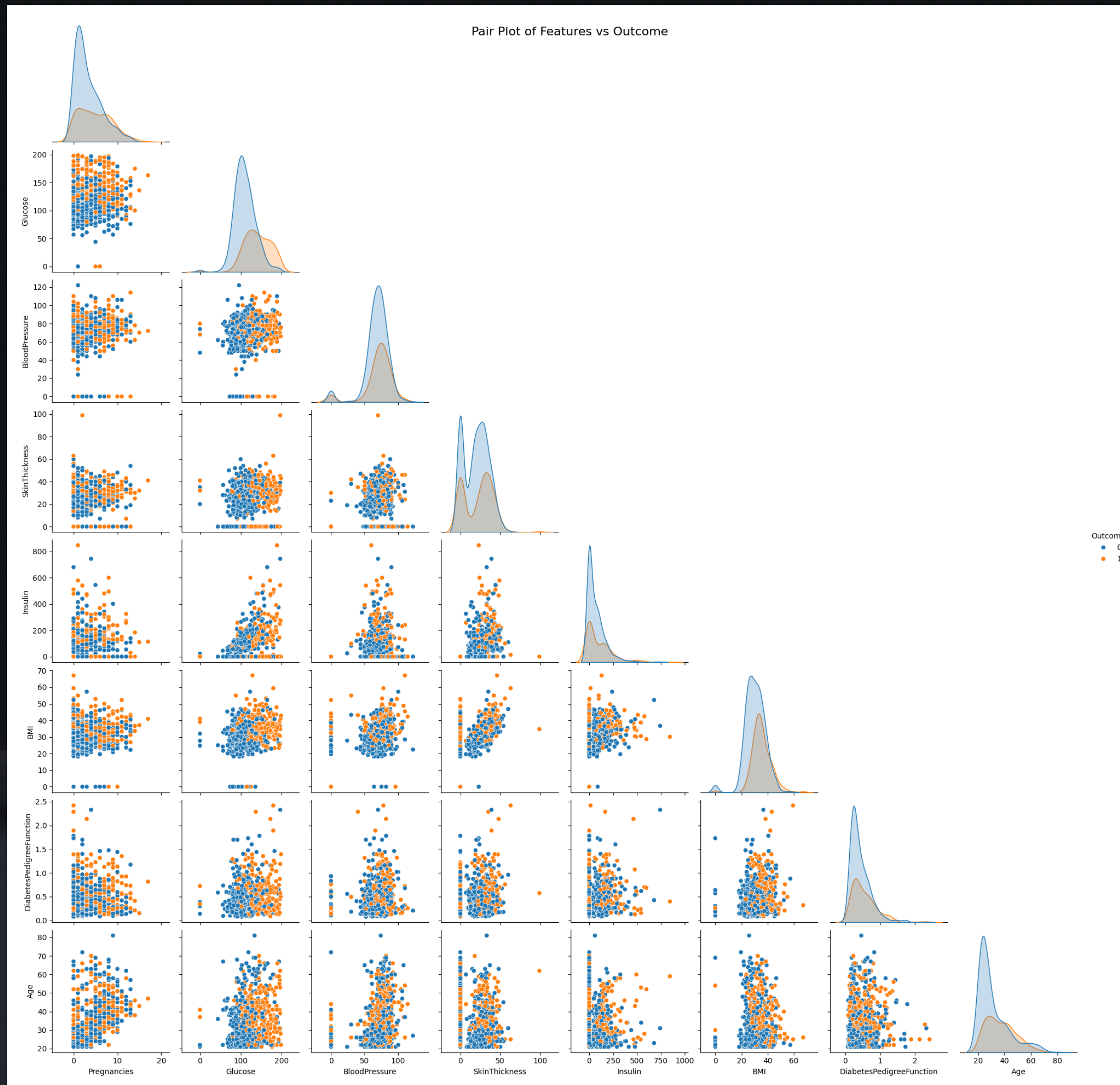
Preprocessing

- *Handel Discretization*
 - Equal Width Binning
 - **Quantile Binning** ✓
 - **K-Means Binning** ✓
 - Round Values to nearest integer

Handle Discretization



Pair Plot

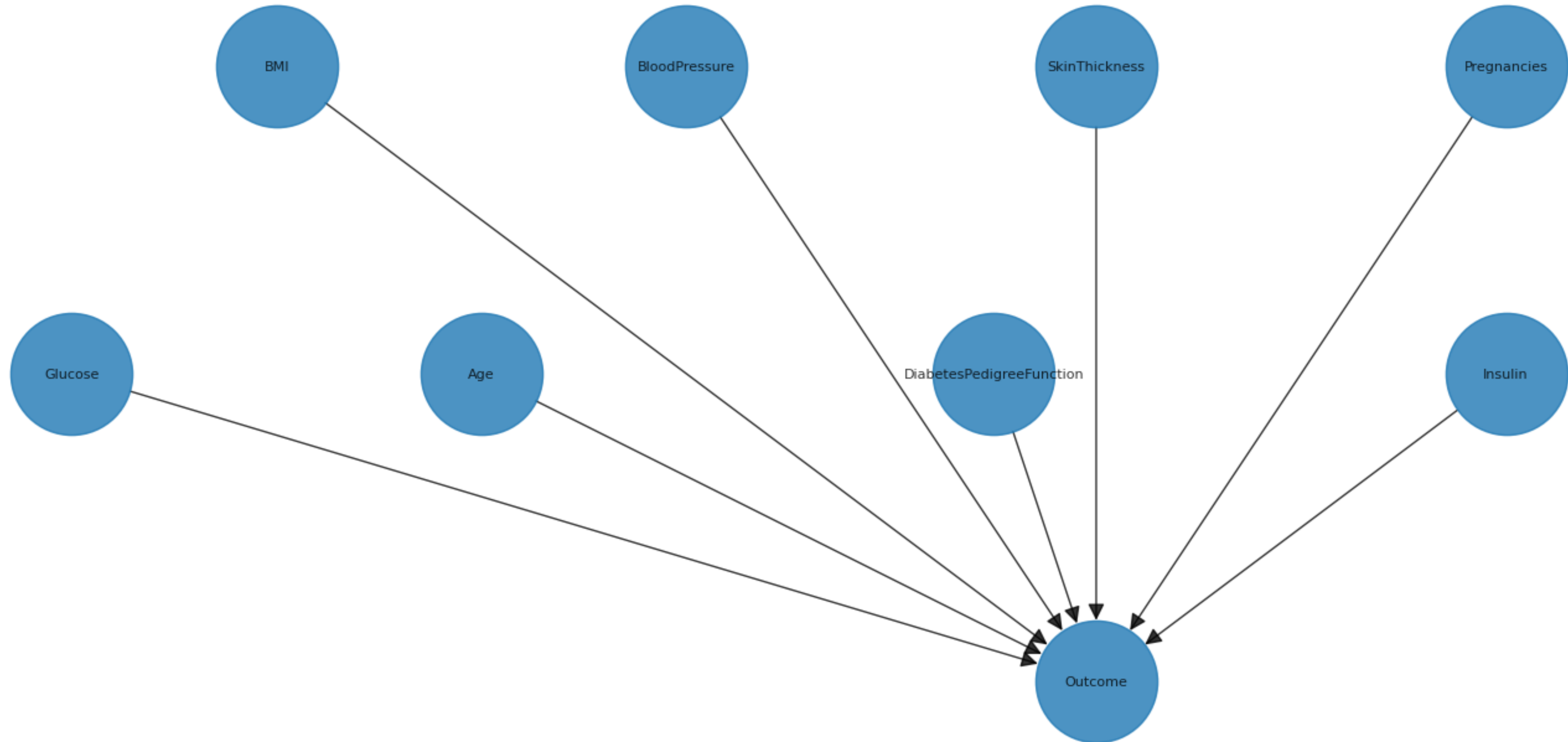


Training Bayesian Network

- 1. Define the structure of the Bayesian Network which there is three options here:*
 - we can define the structure manually based on domain knowledge. Quantile Binning
 - we can use structure learning algorithms to learn the structure from data.
 - we can use a hybrid approach where we define some parts of the
- 2. Find the Conditional Probability Distributions (CPDs) for each node in the network.*
- 3. Perform inference on the Bayesian Network to make analyses with variable elimination. We will perform inferences to analyses the result of different scenarios.*

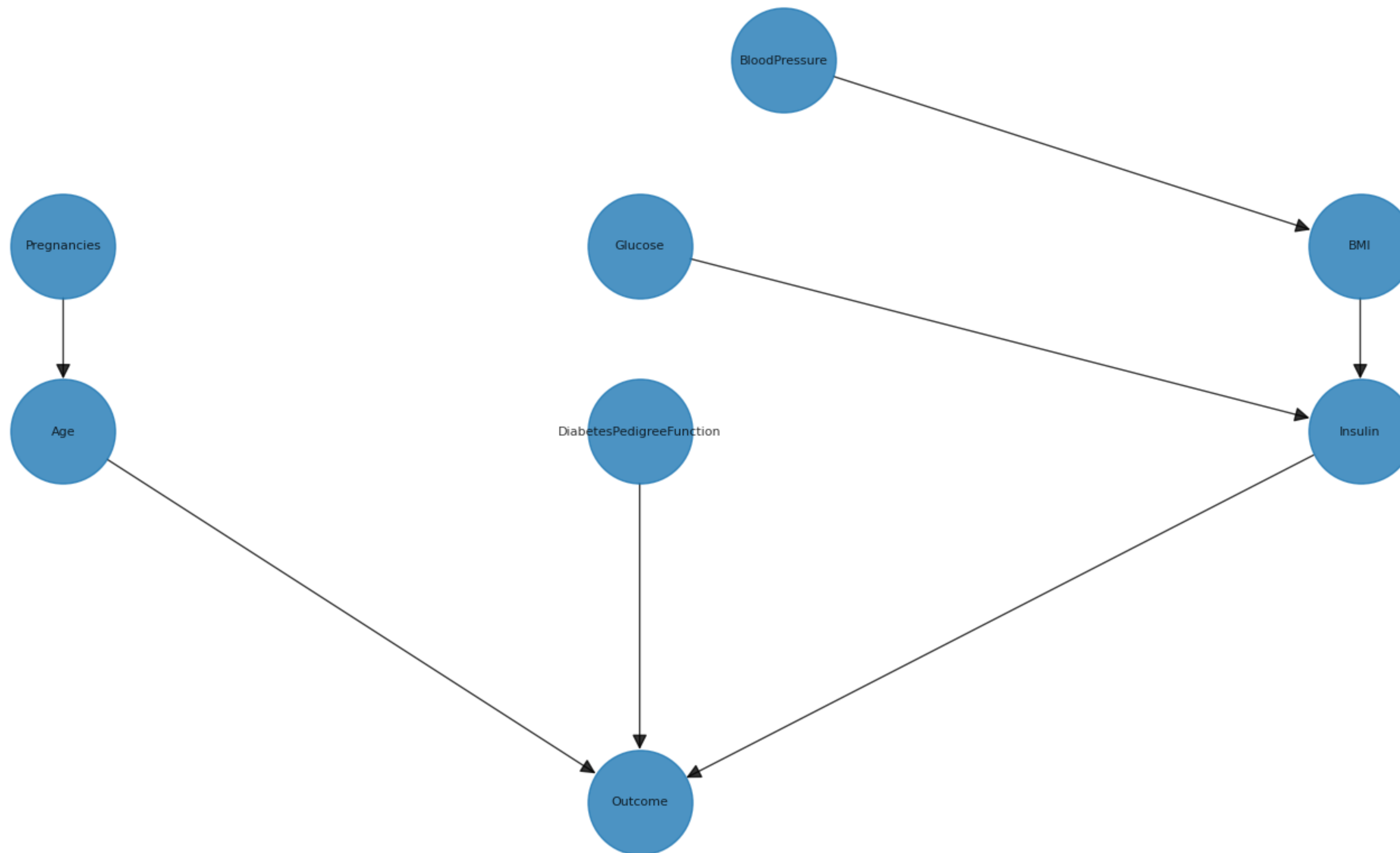
Basic Bayesian Network

Model 1 - Basic Bayesian Network

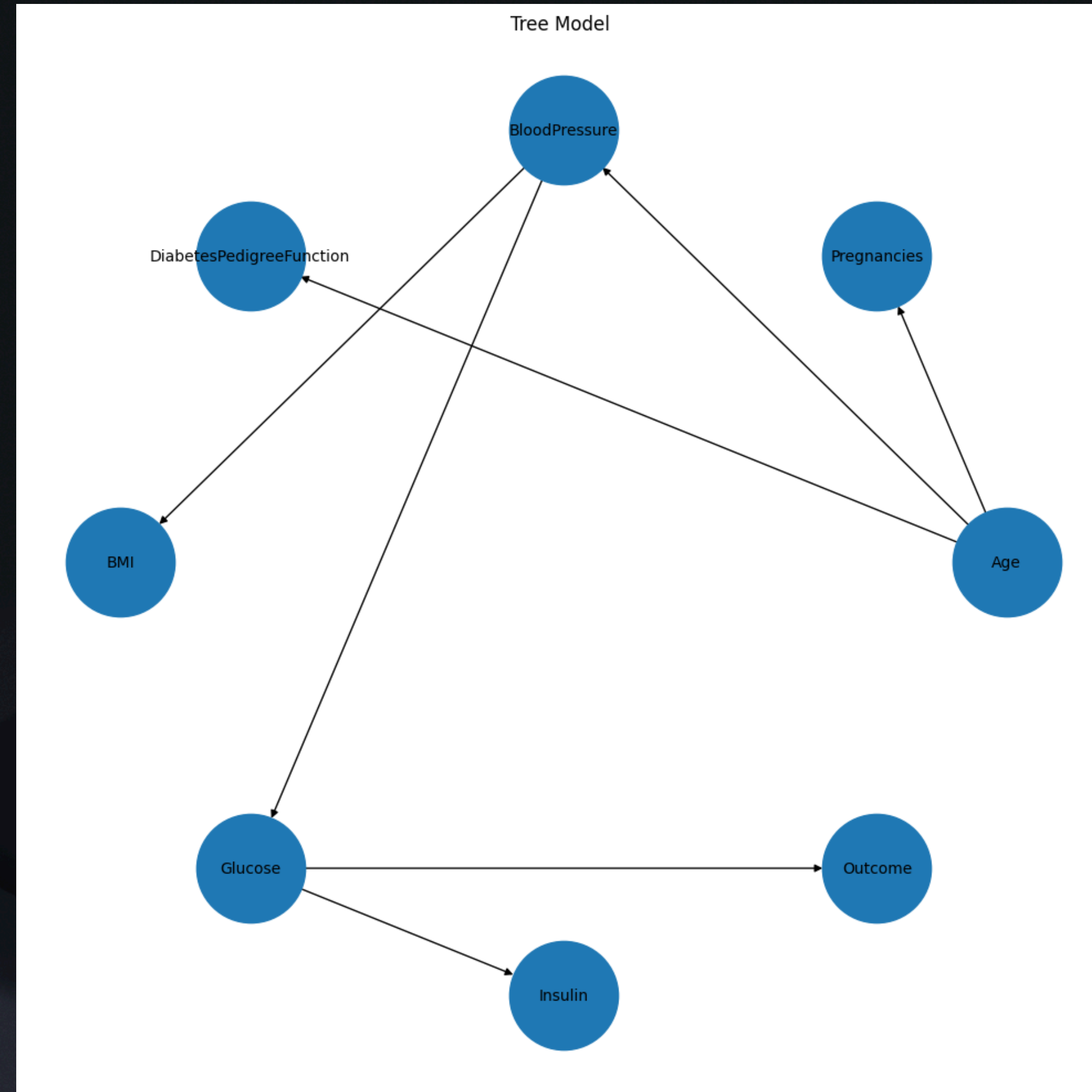


Grouped Feature Model

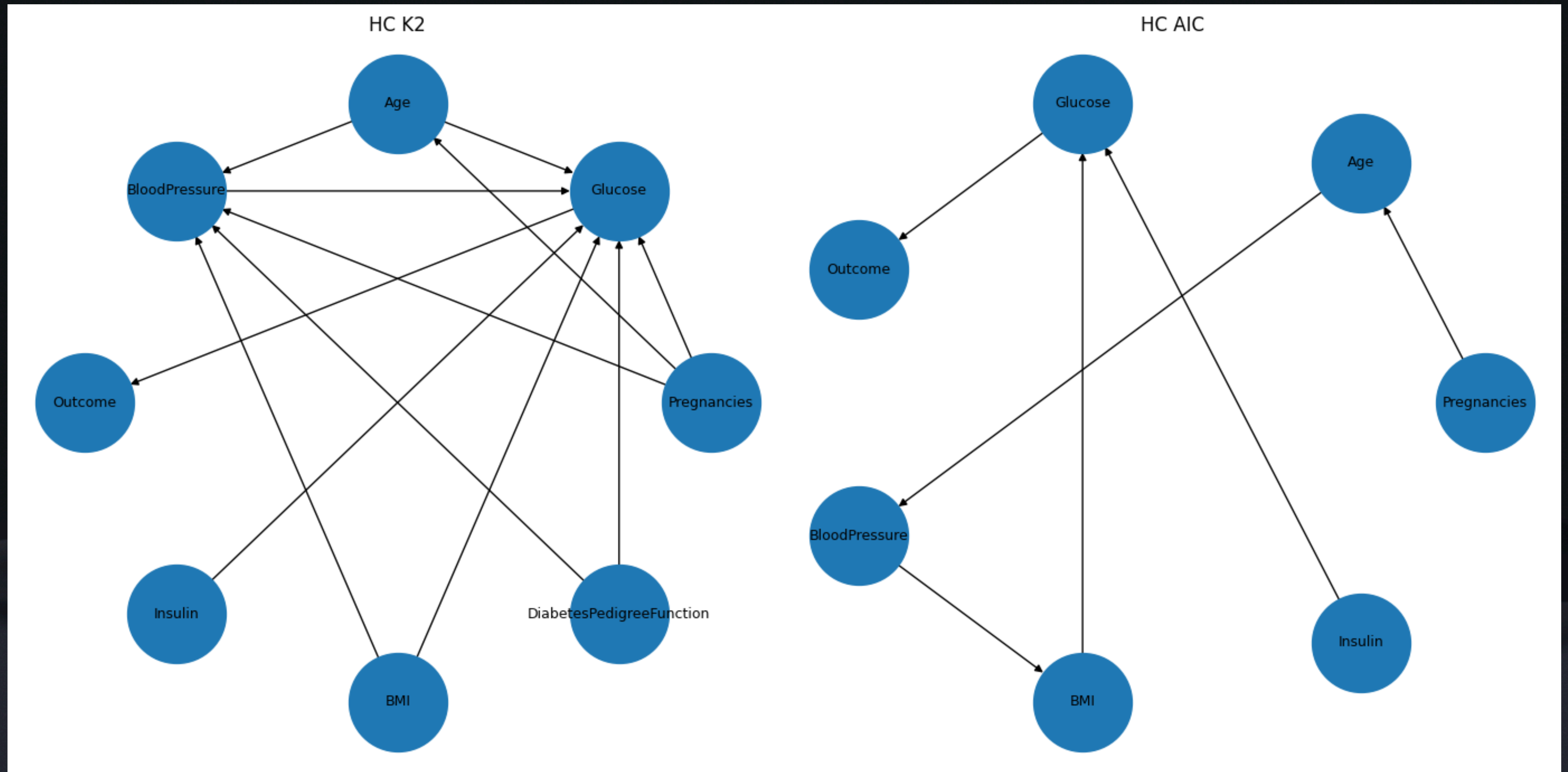
Model 2 - Grouped Feature Bayesian Network



Tree Models



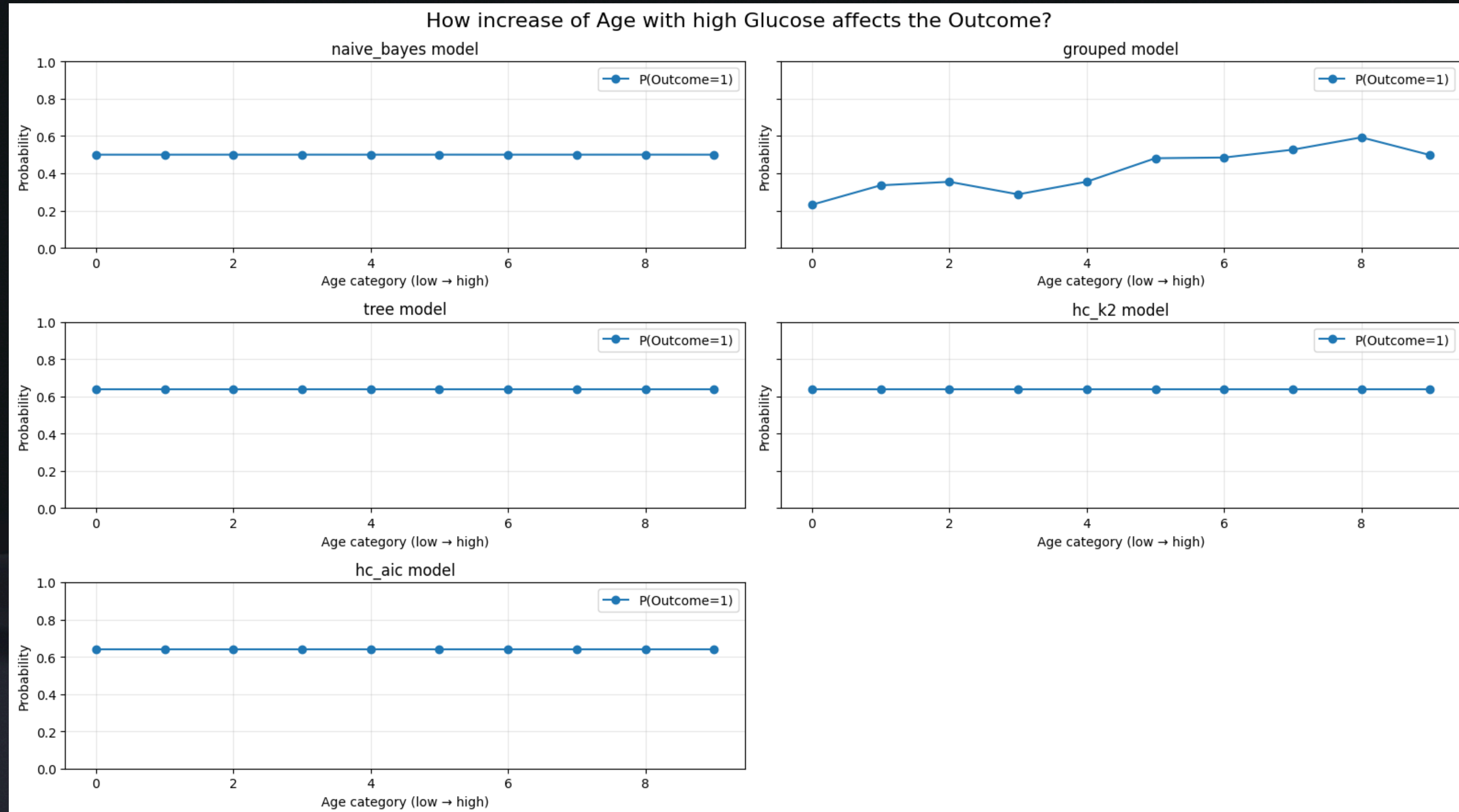
Hill Climbing



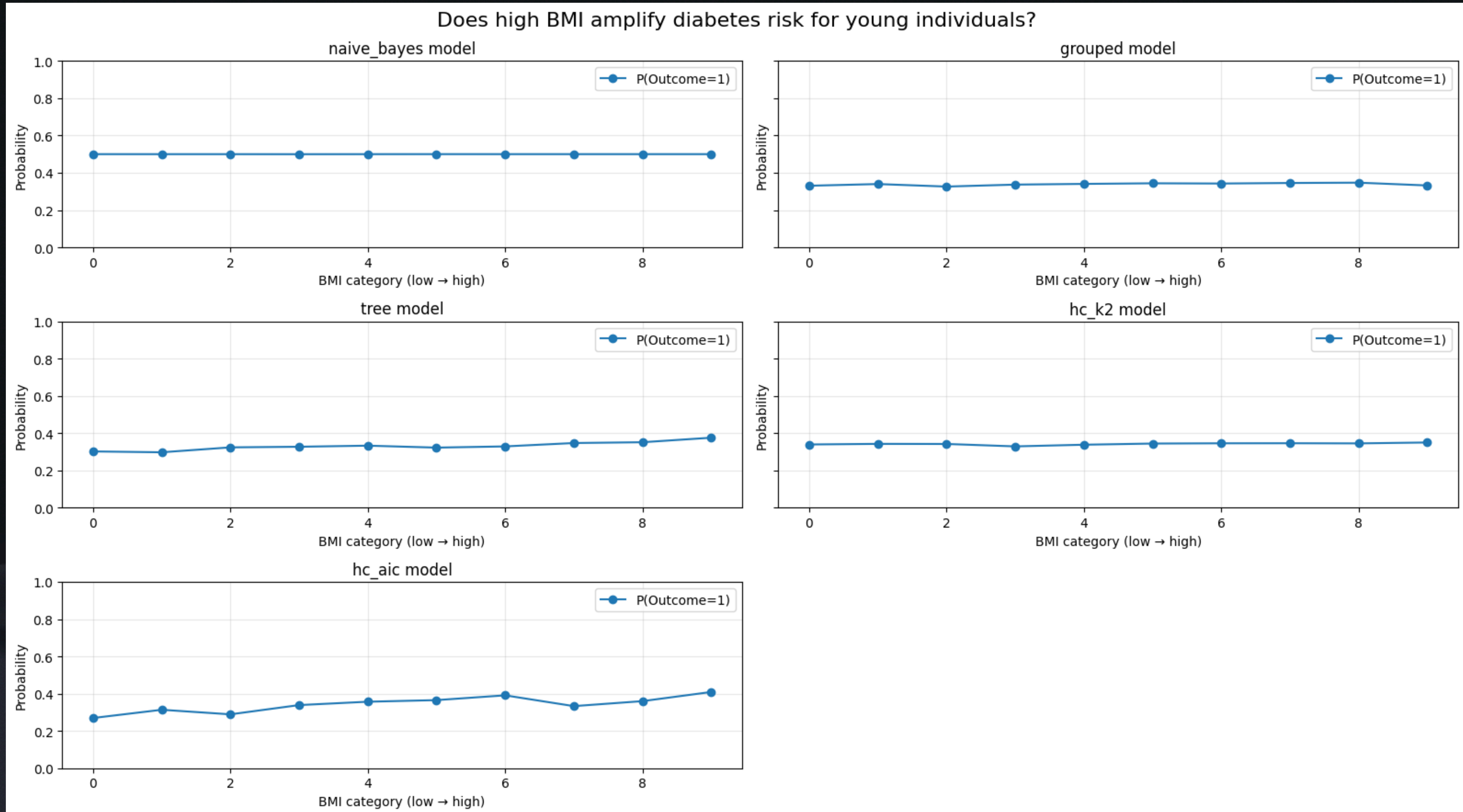
Training

```
Result of model: hc_k2
Independence assertions num: 16
Local semantics of the current model:
(Pregnancies  $\perp$  DiabetesPedigreeFunction, Insulin, BMI)
(Age  $\perp$  DiabetesPedigreeFunction, Insulin, BMI | Pregnancies)
(BloodPressure  $\perp$  Insulin | DiabetesPedigreeFunction, Age, Pregnancies, BMI)
(Outcome  $\perp$  DiabetesPedigreeFunction, Pregnancies, BMI, BloodPressure, Insulin, Age | Glucose)
(Insulin  $\perp$  DiabetesPedigreeFunction, Pregnancies, BMI, BloodPressure, Age)
(BMI  $\perp$  DiabetesPedigreeFunction, Insulin, Age, Pregnancies)
(DiabetesPedigreeFunction  $\perp$  Insulin, Age, Pregnancies, BMI)
Checking Markov blankets
The Markov blanket of node Pregnancies is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Age', 'BMI', 'BloodPressure']
The Markov blanket of node Glucose is ['DiabetesPedigreeFunction', 'Insulin', 'Outcome', 'Age', 'Pregnancies', 'BMI', 'BloodPressure']
The Markov blanket of node Age is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Pregnancies', 'BMI', 'BloodPressure']
The Markov blanket of node BloodPressure is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Age', 'Pregnancies', 'BMI']
The Markov blanket of node Outcome is ['Glucose']
The Markov blanket of node Insulin is ['DiabetesPedigreeFunction', 'Glucose', 'Age', 'Pregnancies', 'BMI', 'BloodPressure']
The Markov blanket of node BMI is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Age', 'Pregnancies', 'BloodPressure']
The Markov blanket of node DiabetesPedigreeFunction is ['Glucose', 'Insulin', 'Age', 'Pregnancies', 'BMI', 'BloodPressure']
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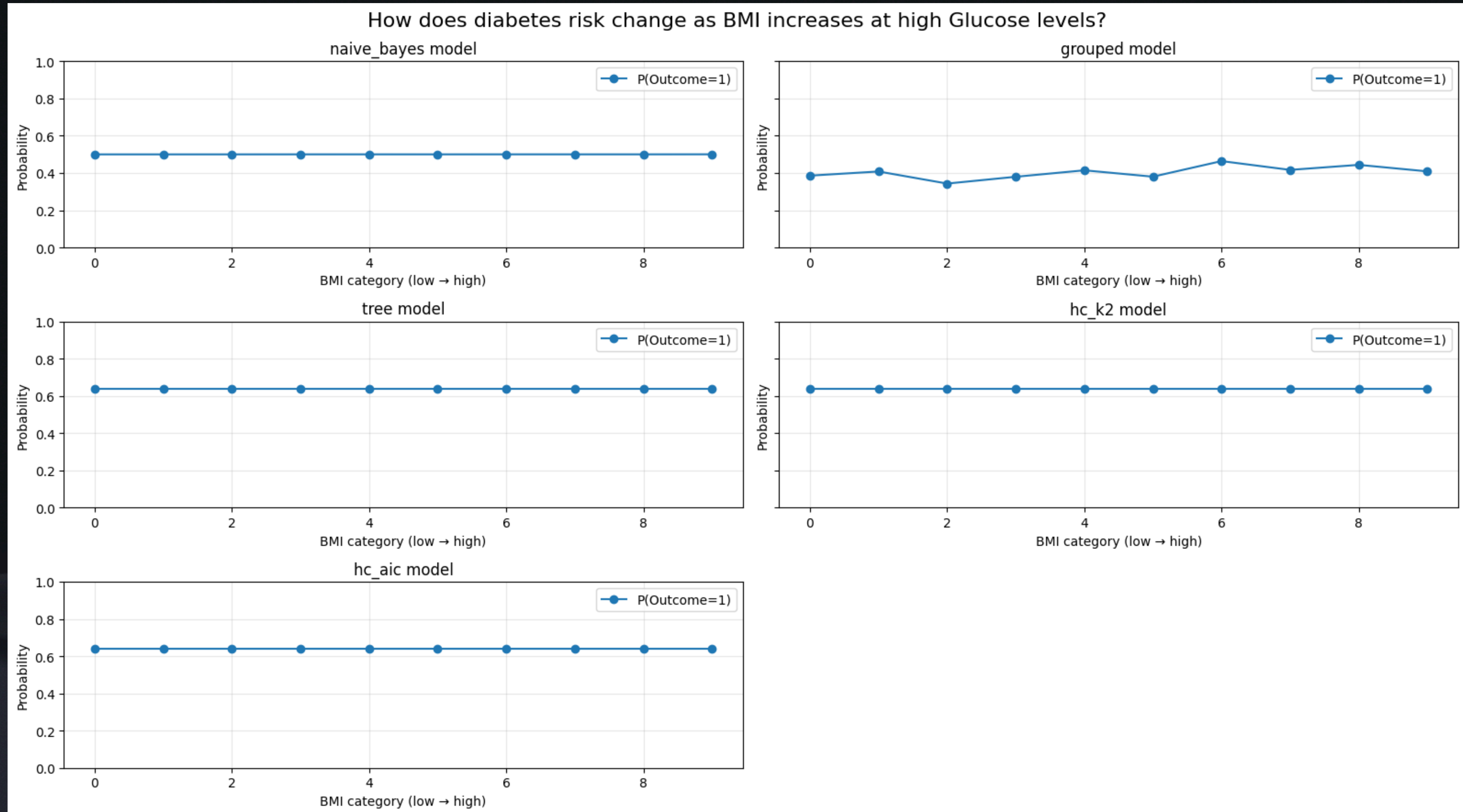

Question 1: How does diabetes risk change when Age increases while Glucose remains high?



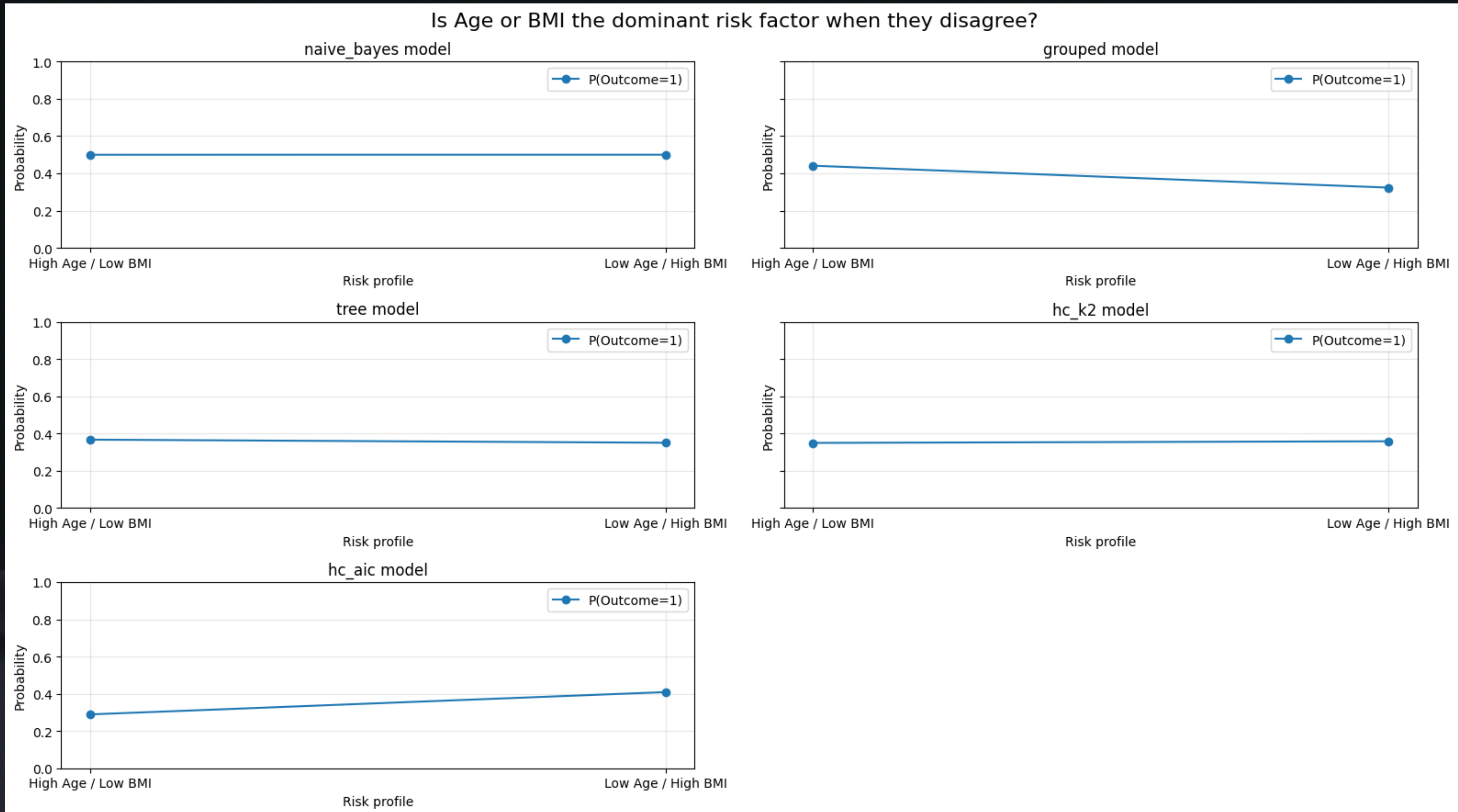
Question 2: Does high BMI amplify diabetes risk for young individuals?



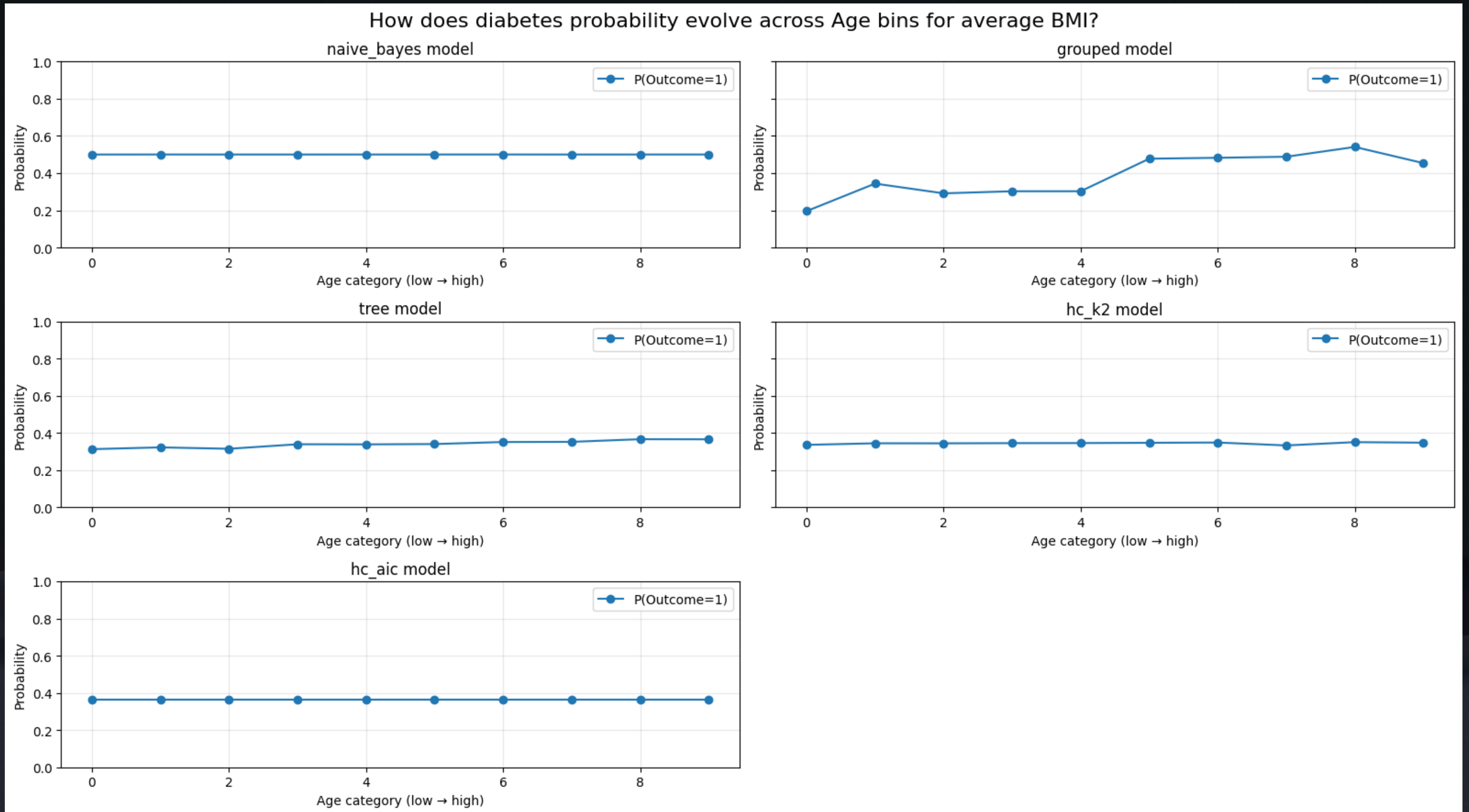
Question 3: Can low BMI reduce diabetes risk even at high Glucose levels?



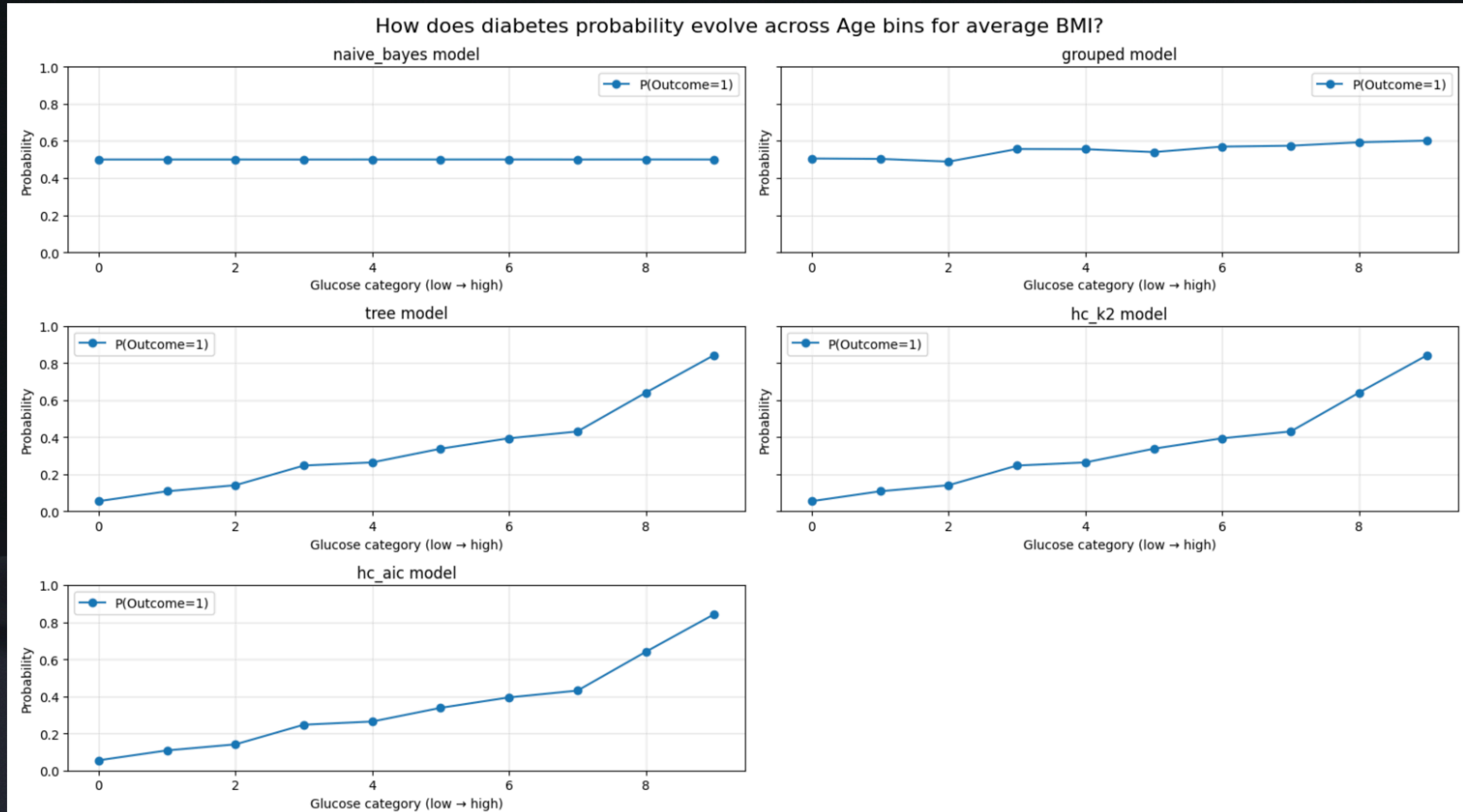
Question 4: Is Age or BMI the dominant risk factor when they disagree?



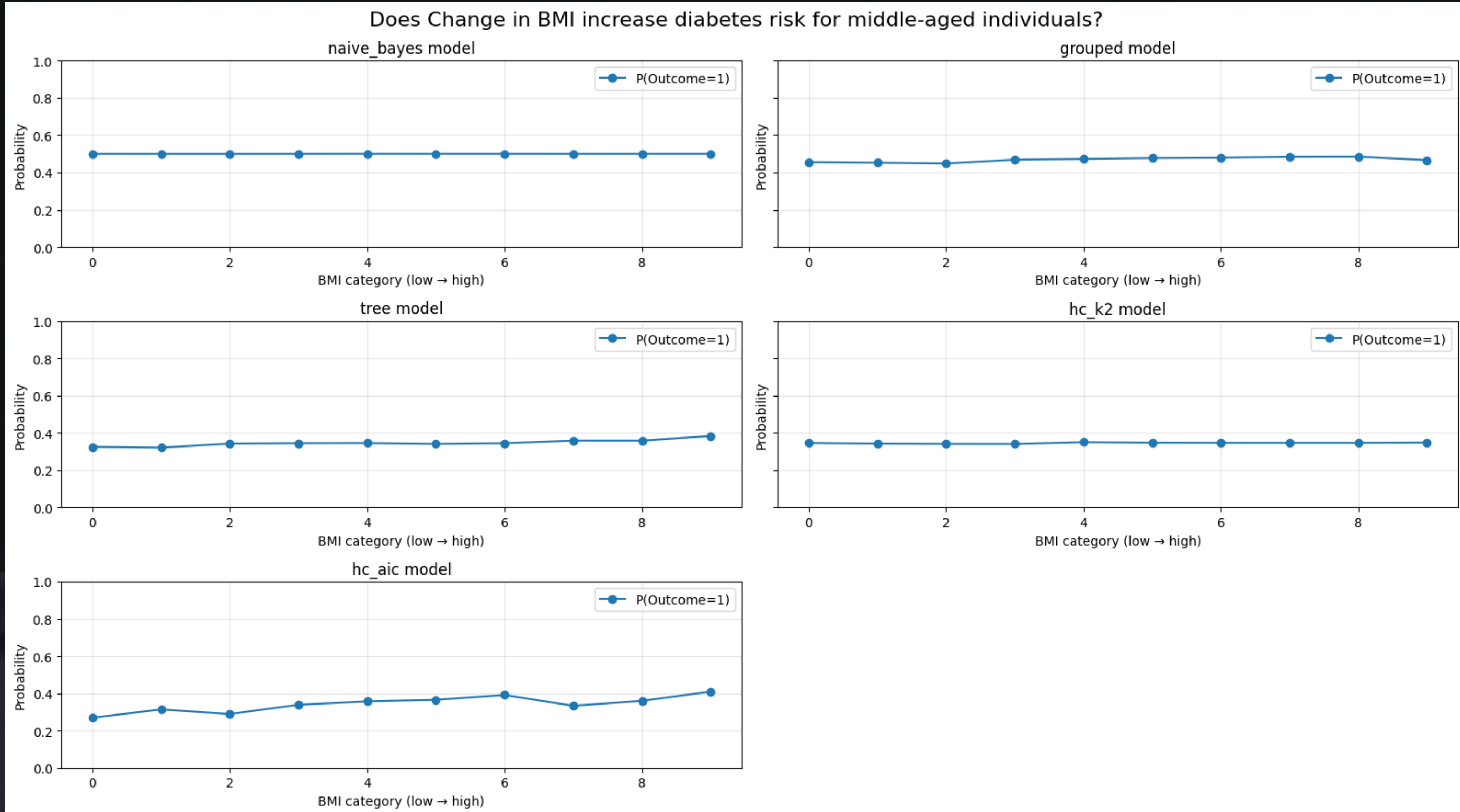
Question 5: How does diabetes probability evolve across Age bins for average BMI?



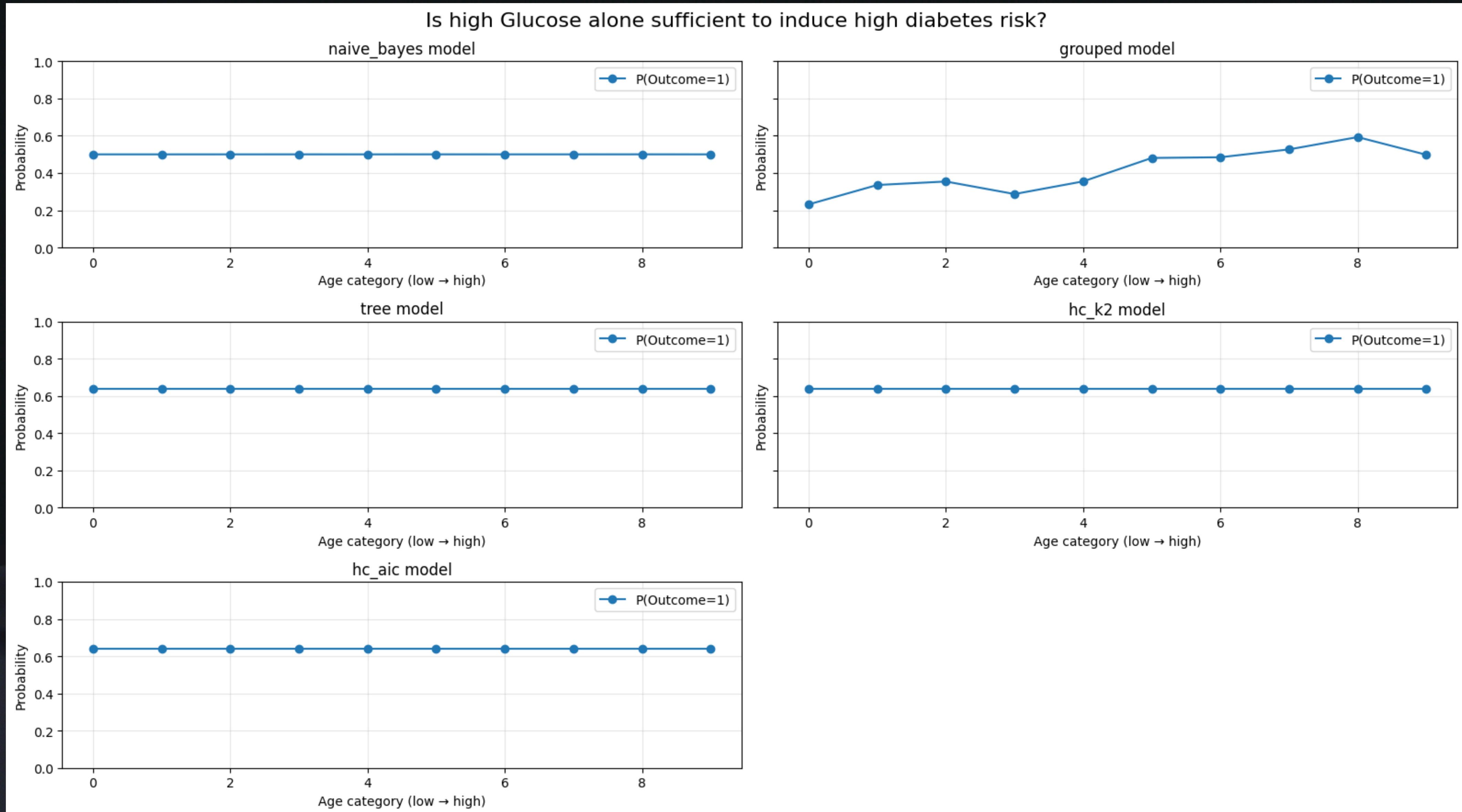
Question 6: At what Glucose level does diabetes risk exceed 50% for elderly individuals?



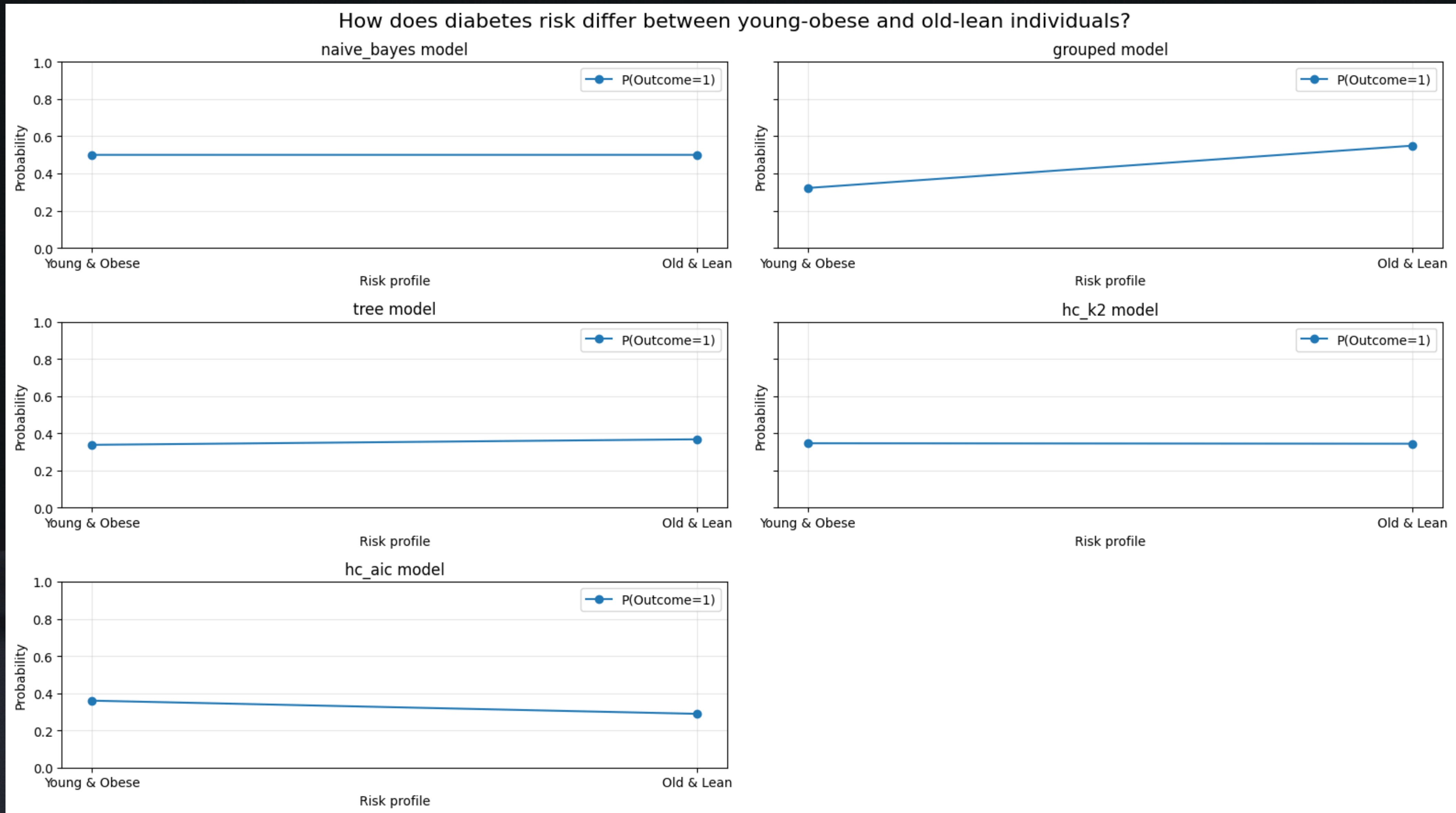
Question 7: Does Change in BMI increase diabetes risk for middle-aged individuals?



Question 8: Is high Glucose alone sufficient to induce high diabetes risk?

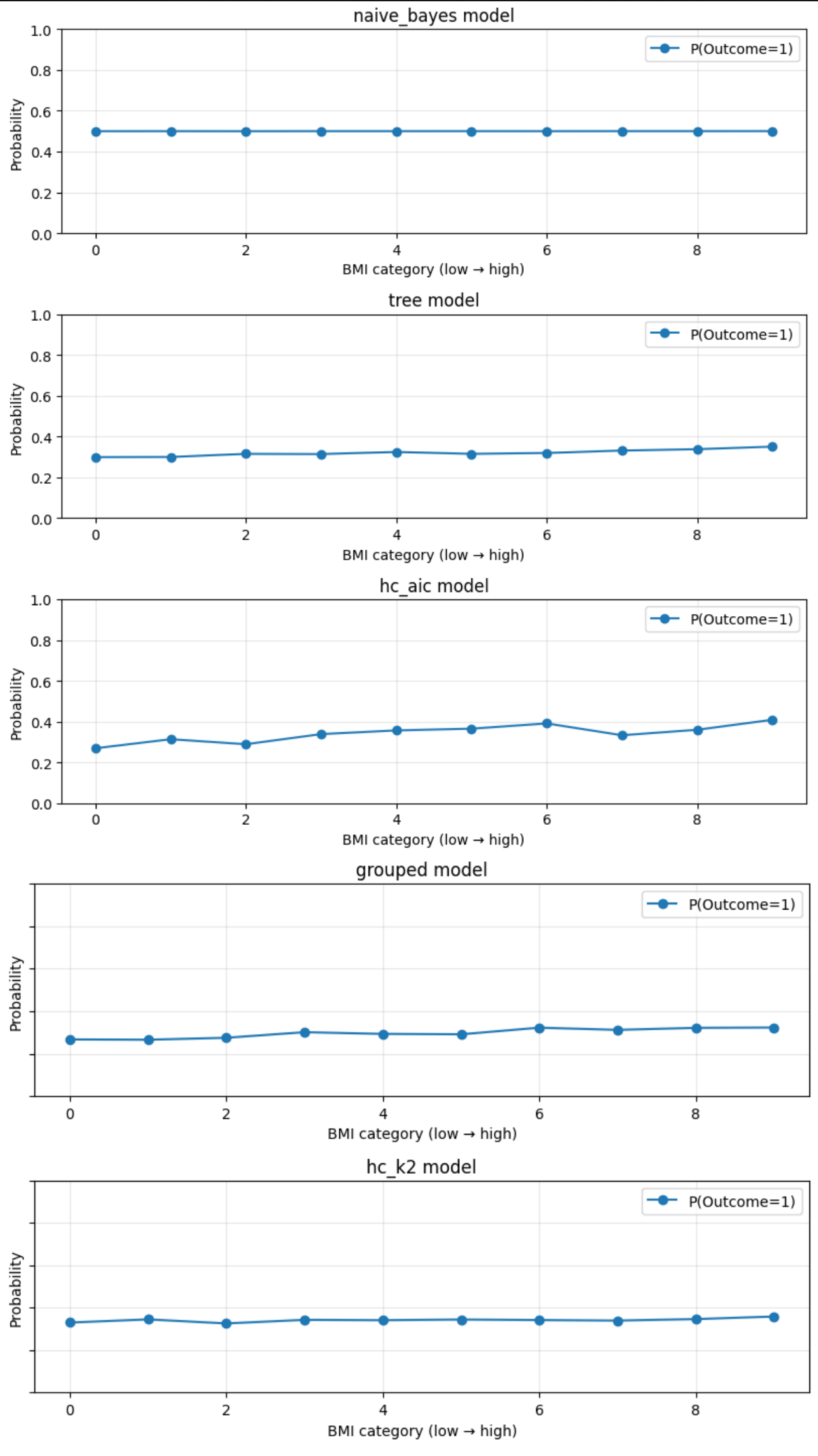


Question 9: How does diabetes risk differ between young-obese and old-lean individuals?

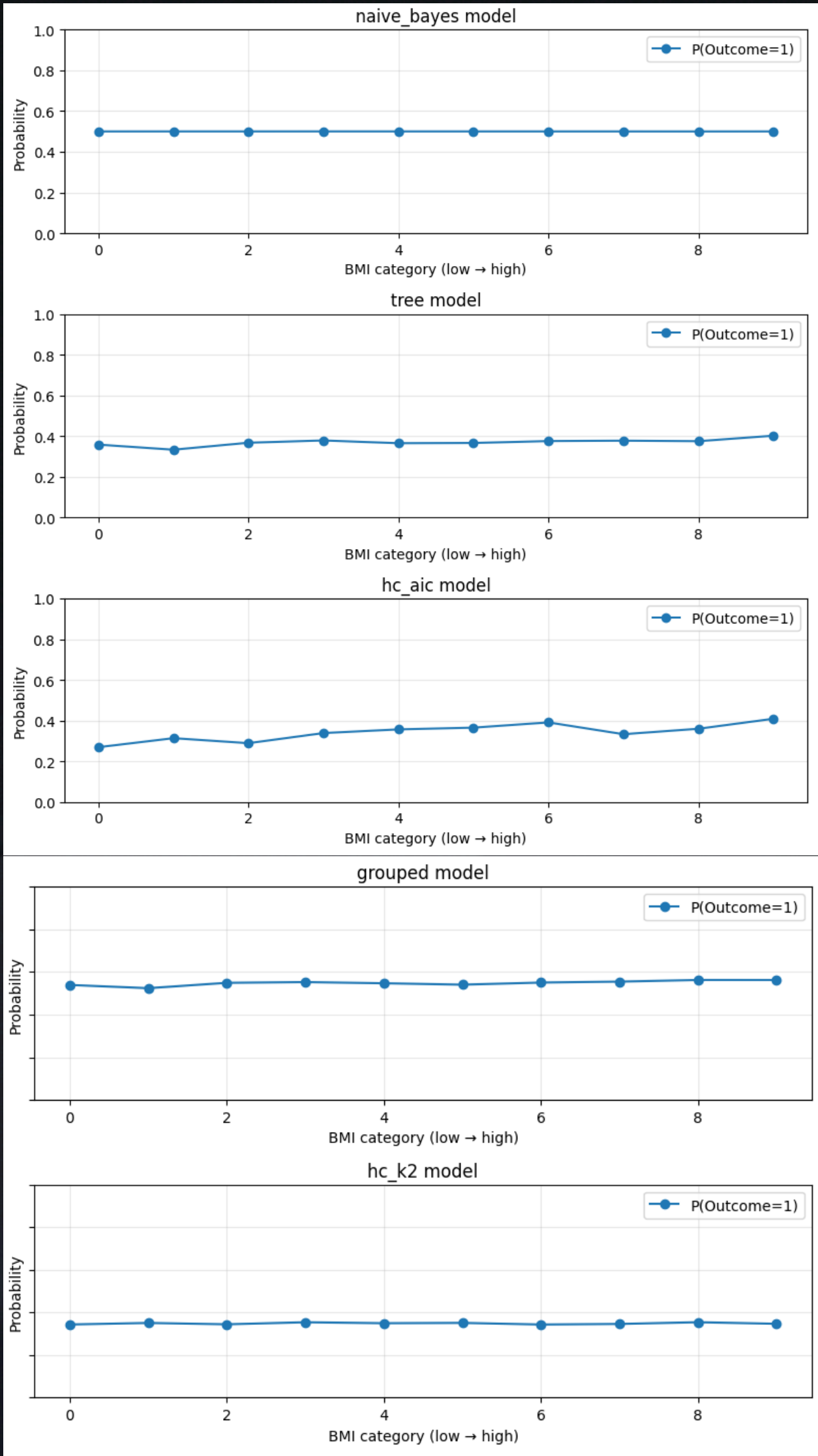


Question 10: Which age group shows the highest sensitivity to BMI changes?

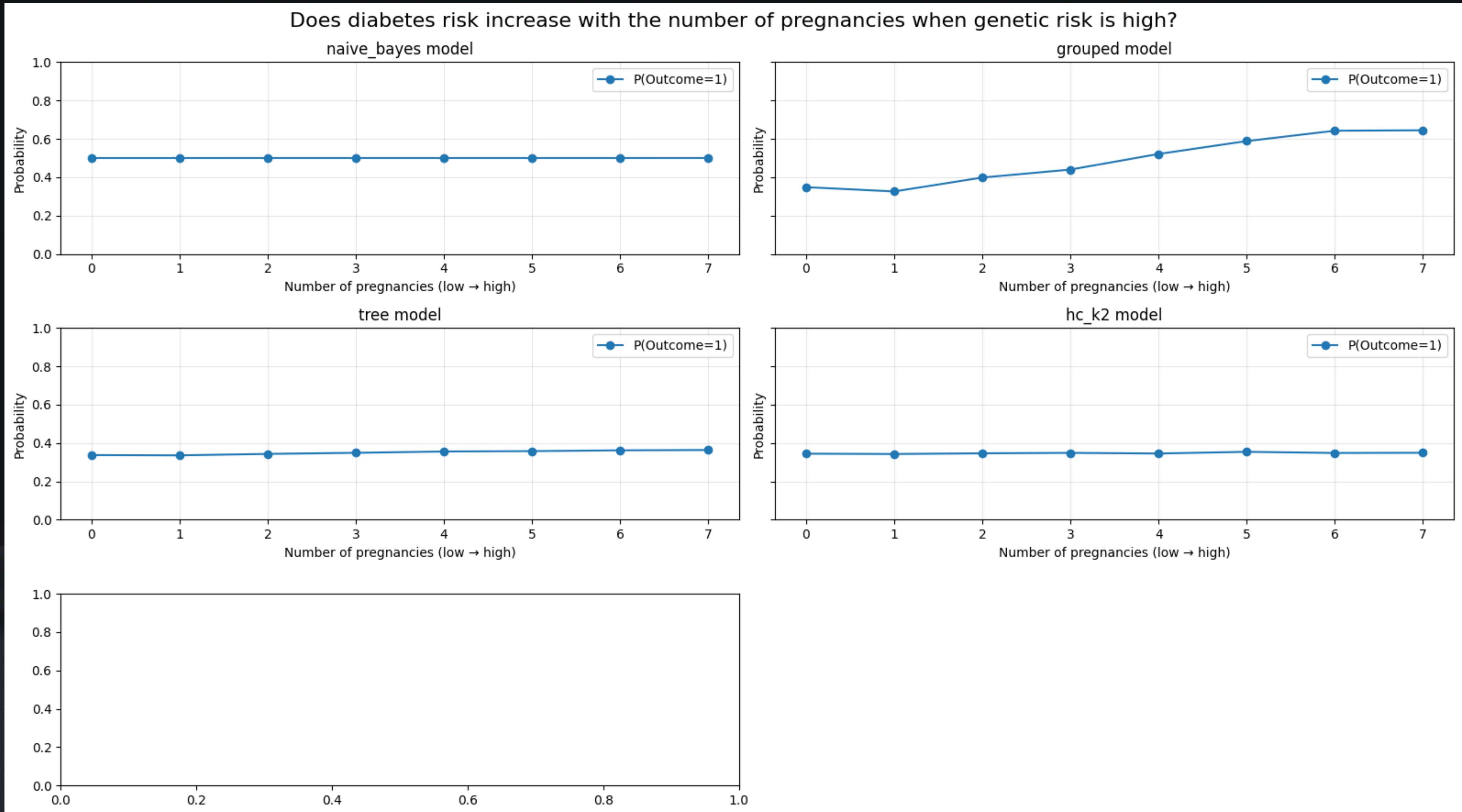
BMI sensitivity for young individuals



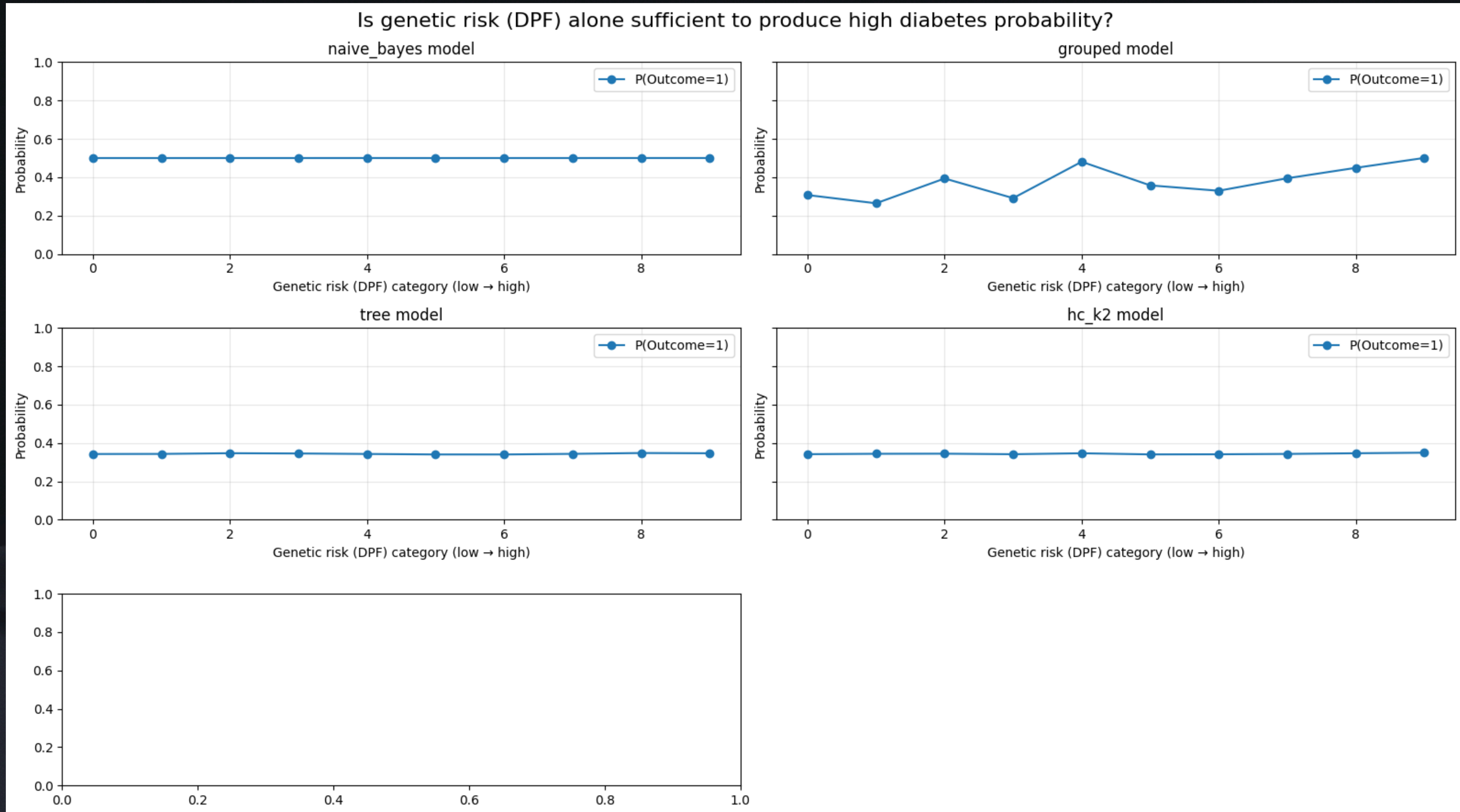
BMI sensitivity for old individuals



Question 11 : Does diabetes risk increase with the number of pregnancies when genetic risk is high?



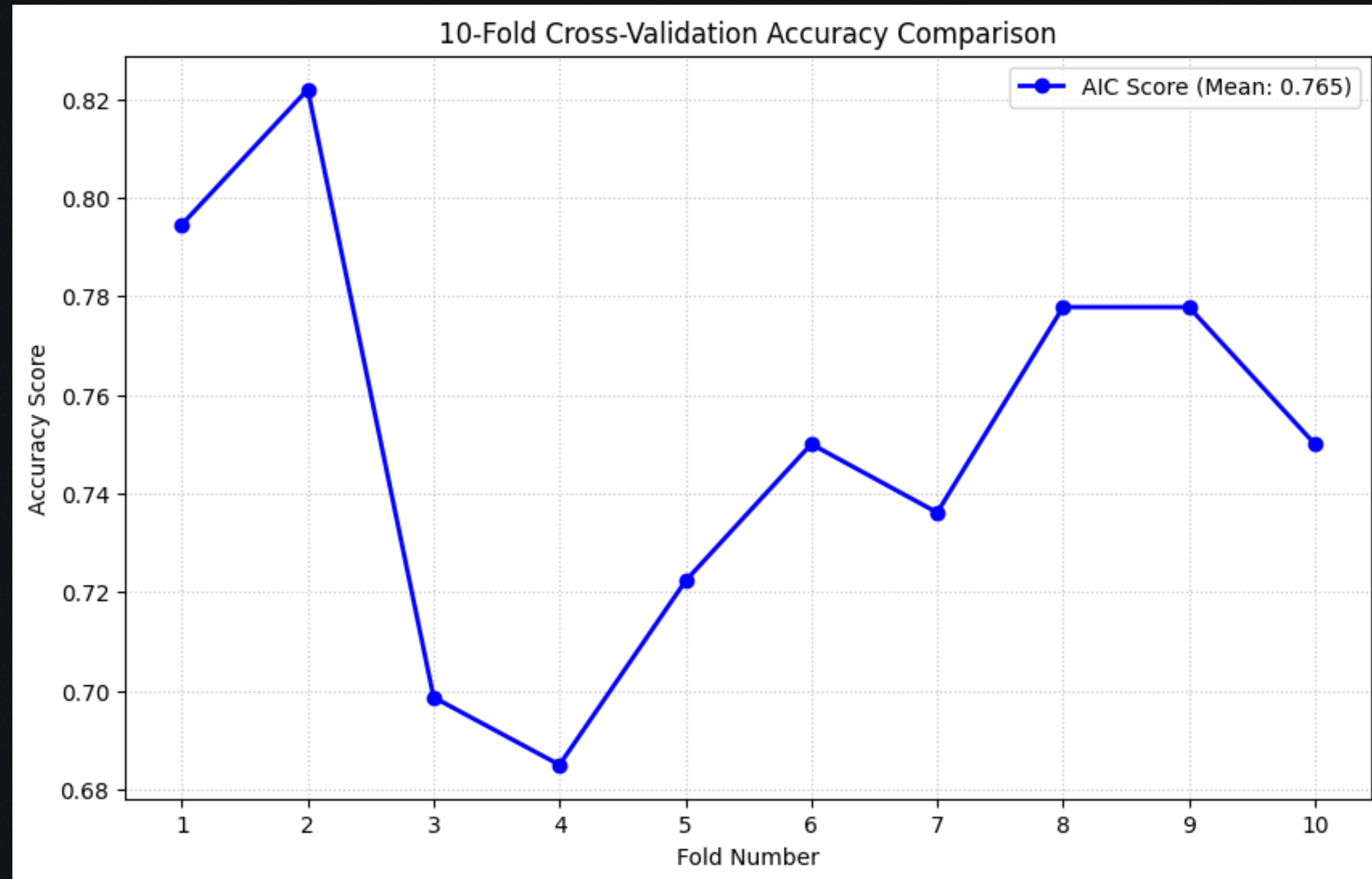
Question 12: Is genetic risk (DPF) alone sufficient to produce high diabetes probability?



Comparative Performance Table

Model	Sensitivity	Structural Integrity	Clinical Utility	Verdict
Naive Bayes	Very Low	Low	Very Low	Failure: Cannot propagate evidence.
Grouped	Low	Medium	Low	Insufficient: Too rigid to capture data nuances.
Tree	High	Medium	Medium	Informative: Useful but potentially over-simplistic.
HC (K2)	Very High	High	Medium	Aggressive: Powerful but prone to overconfidence.
HC (AIC)	High	High	High	Optimal: Best balance of complexity and accuracy.

Bayesian Network Classification using Hill Climbing



Thank You